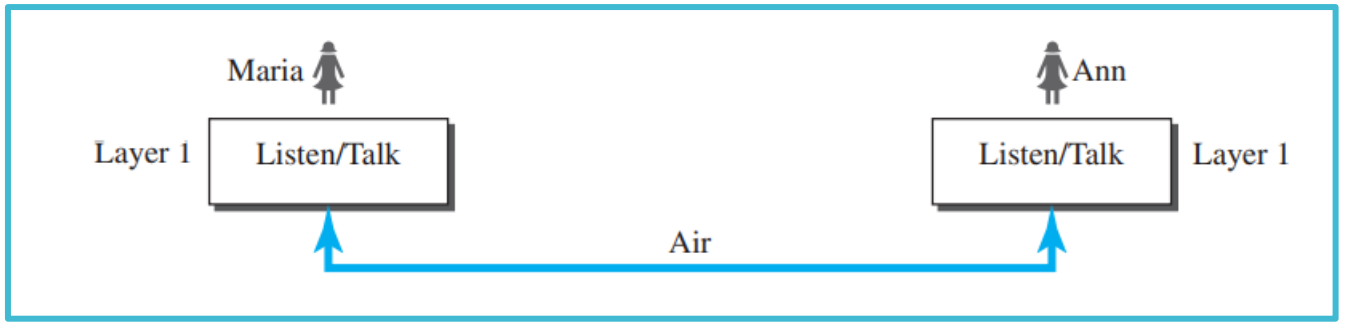
PROTOCOL LAYERING

Definition

- When communication is simple - need only one simple protocol.
- when the communication is complex - need to divide the task between different layers, in which case we need a protocol at each layer, or **protocol layering**.

Need for Layered Protocol

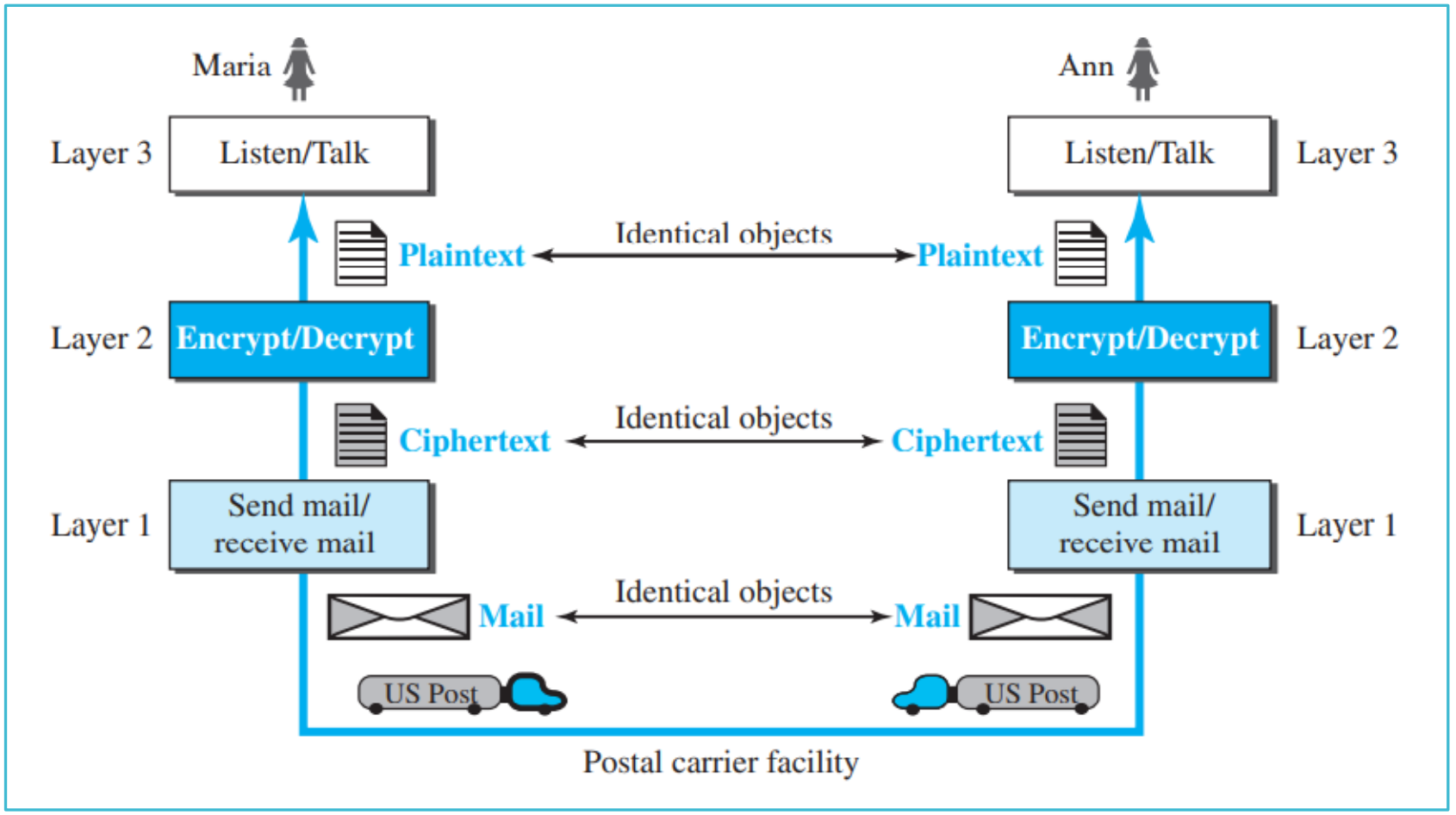
Single Layer Protocol:
Communication is so simple that it can occur in only one layer



Single Layer Protocol

- First, Maria and Ann know that they should **greet each other when they meet**.
- Second, they know that they should confine their vocabulary to the level of their friendship.
- Third, each party knows that she should refrain from speaking when the other party is speaking.
- Fourth, each party knows that the conversation should be a dialog, not a monolog: both should have the opportunity to talk about the issue.
- Fifth, they should exchange some nice words when they leave

Three Layer Protocol



Principles of Protocol Layering

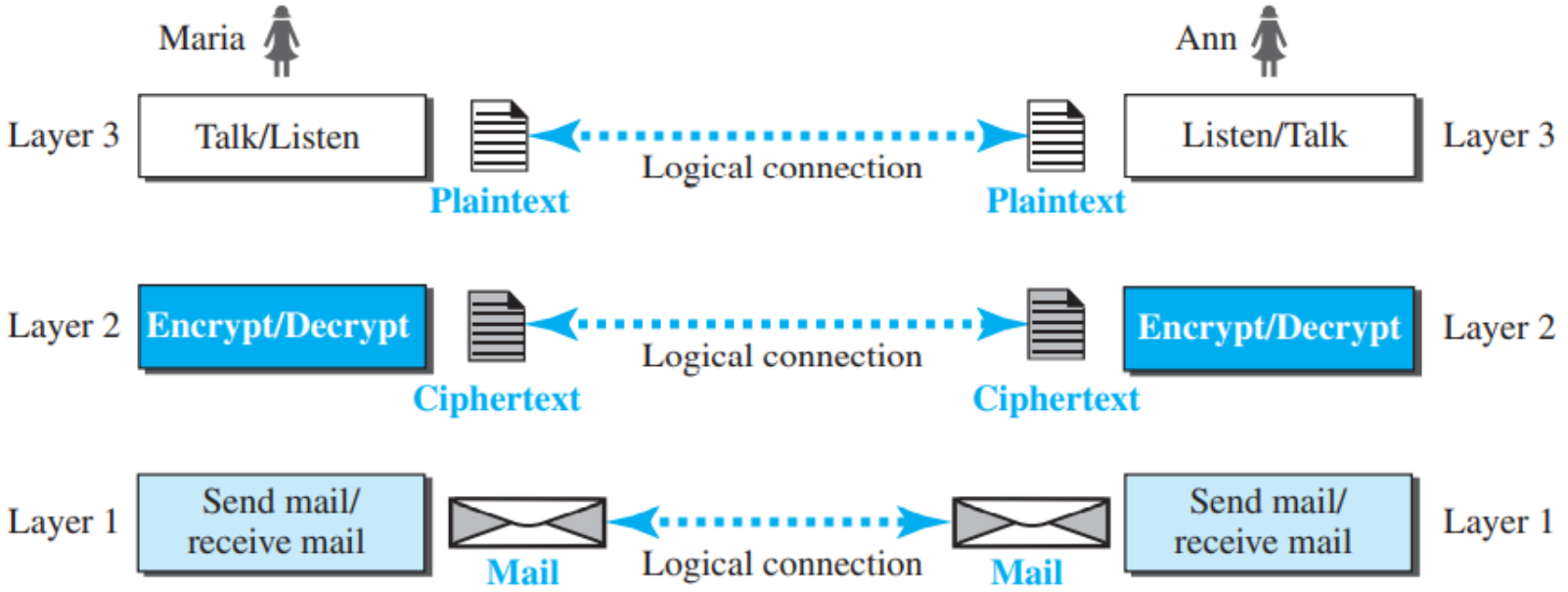
- The first principle - bidirectional communication, we need to make each layer so that it is able to perform two opposite tasks, one in each direction.
- For example, the third layer task is to listen (in one direction) and talk (in the other direction).
- The second layer needs to be able to encrypt and decrypt.
- The first layer needs to send and receive mail.

Principles of Protocol Layering

- The second principle - the two objects under each layer at both sites should be identical.
- For example, the object under layer 3 at both sites should be a plaintext letter.
- The object under layer 2 at both sites should be a ciphertext letter.
- The object under layer 1 at both sites should be a piece of mail.

Logical Connection

Figure 2.3 *Logical connection between peer layers*



TCP/IP PROTOCOL SUITE

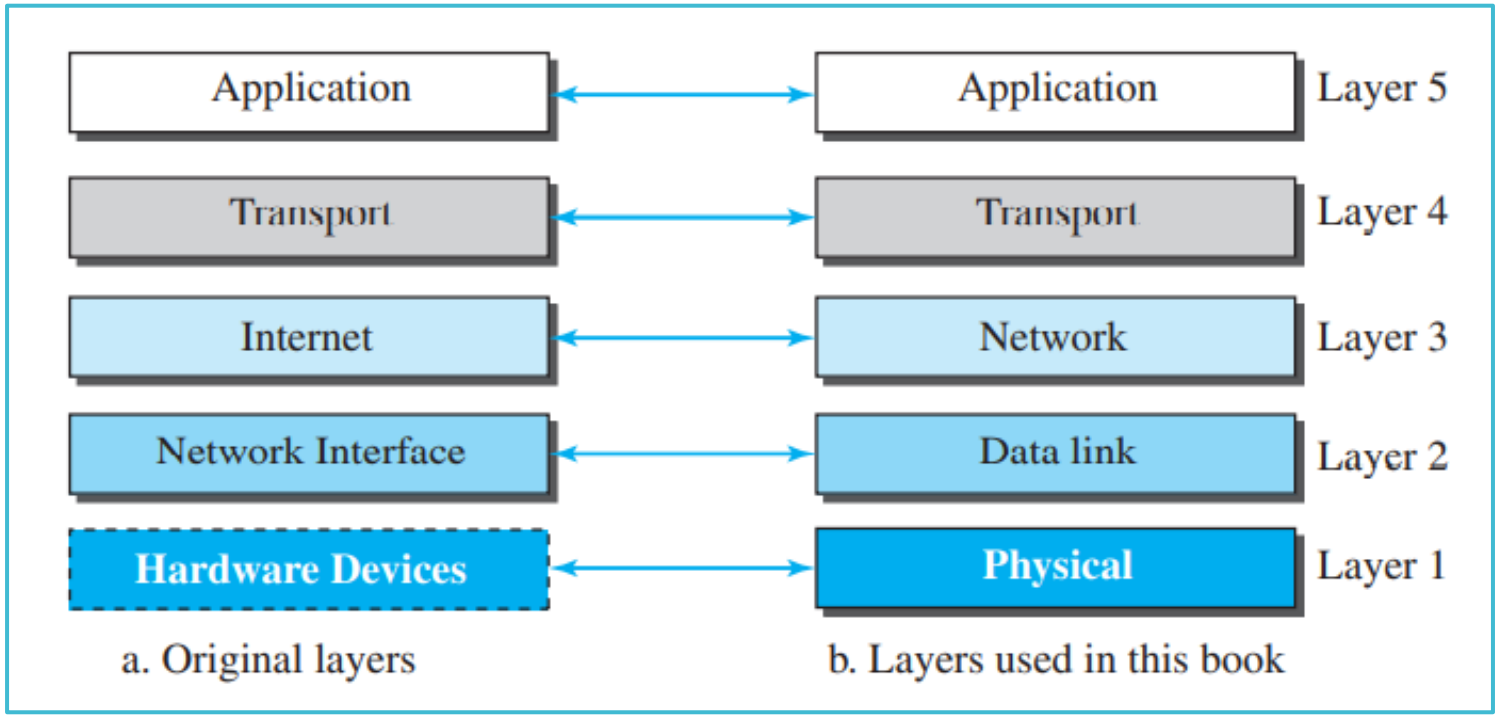
Contd…

- The **TCP/IP (Transmission Control Protocol/ Internet Protocol).**
- TCP/IP is a protocol suite (a set of protocols organized in different layers) used in the Internet today.
- It is a hierarchical protocol made up of interactive modules, each of which provides a specific functionality.

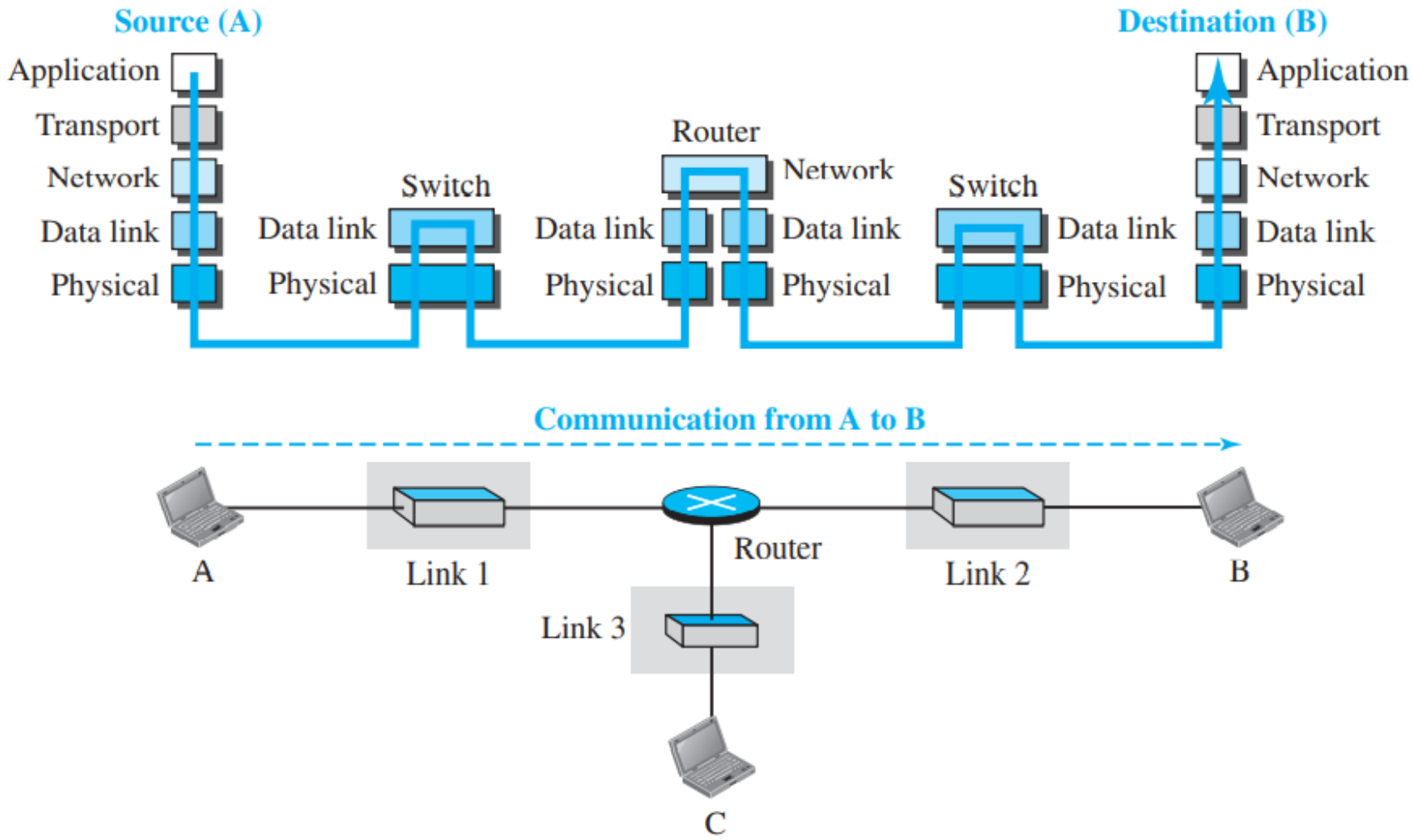
Contd…

- The term hierarchical means that each upper-level protocol is supported by the services provided by one or more lower-level protocols.
- The original TCP/IP protocol suite was defined as four software layers built upon the hardware.
- TCP/IP is thought of as a five-layer model.

Layered Architecture



Communication through an internet



Contd…

- Five communicating devices in this communication:
  1. Source host (computer A)
  2. The link-layer switch in link 1
  3. The router
  4. The link-layer switch in link 2
  5. The destination host (computer B)
- Each device is involved with a set of layers depending on the role of the device in the internet.

Contd…

- The two hosts are involved in all five layers;
- The source host needs to create a message in the application layer and send it down the layers so that it is physically sent to the destination host.
- The destination host needs to receive the communication at the physical layer and then deliver it through the other layers to the application layer.

Contd...

- The router is involved in only three layers;
- There is no transport or application layer in a router as long as the router is used only for routing.
- Although a router is always involved in one network layer, it is involved in n combinations of link and physical layers in which n is the number of links the router is connected to.
- The reason is that each link may use its own data-link or physical protocol.

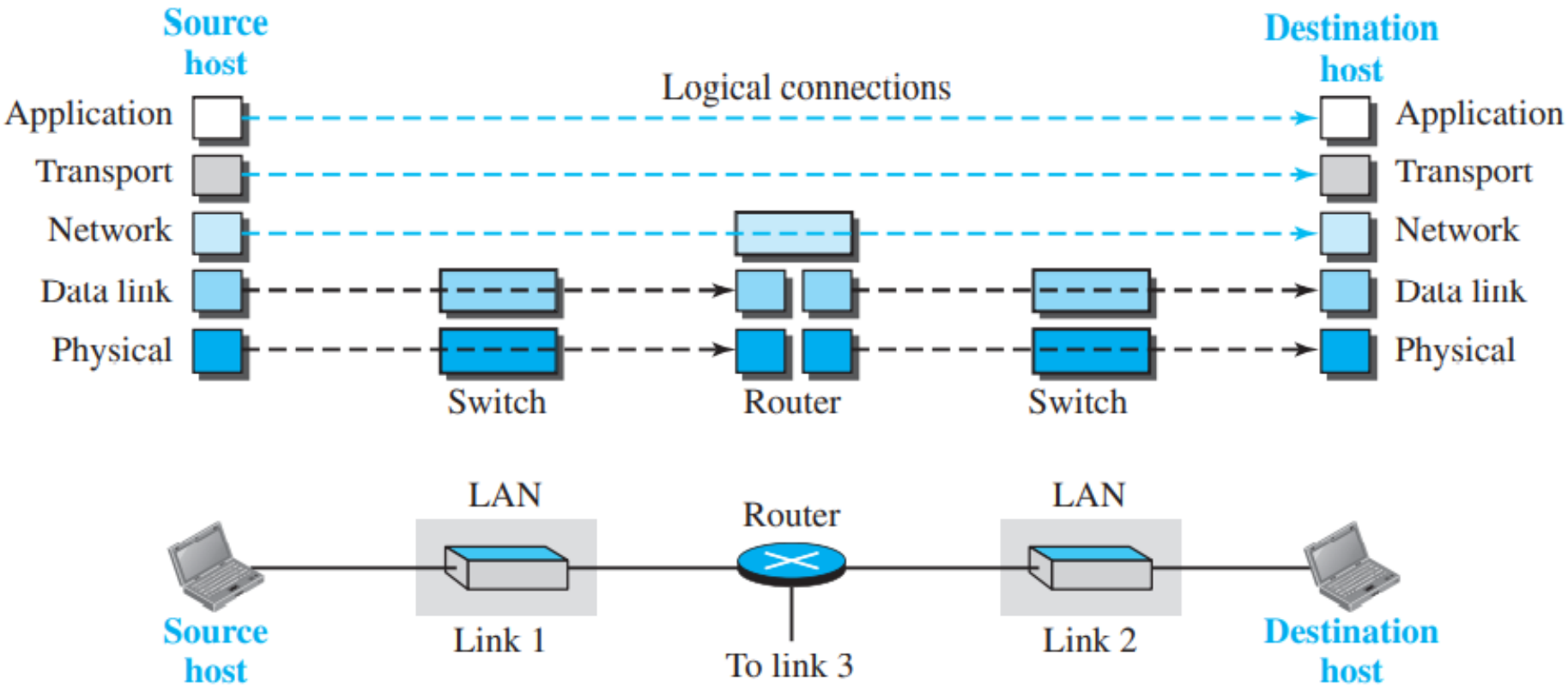
Contd...

- A link-layer switch in a link, however, is involved only in two layers, data-link and physical.
- A router, a link-layer switch is involved only in one data-link and one physical layer.

Layers in the TCP/IP Protocol Suite

- Using logical connections makes it easier for us to think about the duty of each layer.
- The duty of the application, transport, and network layers is end-to-end.
- The duty of the data-link and physical layers is hop-to-hop, in which a hop is a host or router.
- The domain of duty of the top three layers is the internet, and the domain of duty of the two lower layers is the link.

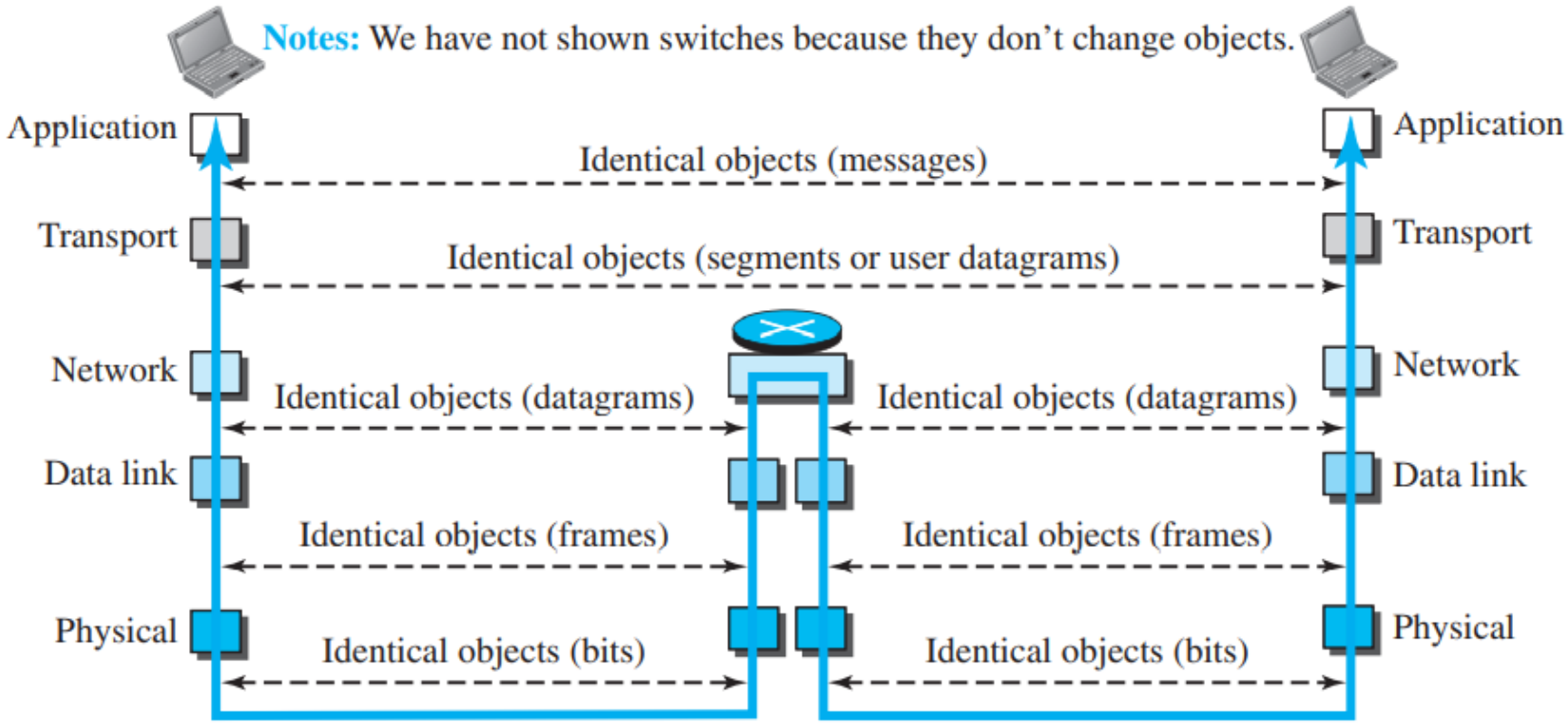
Figure 2.6 *Logical connections between layers of the TCP/IP protocol suite*



Contd…

- In the top three layers, the data unit (packets) should not be changed by any router or link-layer switch.
- In the bottom two layers, the packet created by the host is changed only by the routers, not by the link-layer switches.

Figure 2.7 *Identical objects in the TCP/IP protocol suite*



Contd…

- The logical connection at the network layer is between the two hosts.
- Identical objects exist between two hops in this case because a router may fragment the packet at the network layer and send more packets than received.

Description of Each Layer

- The physical layer is responsible for carrying individual bits in a frame across the link.
- Although the physical layer is the lowest level in the TCP/IP protocol suite, the communication between two devices at the physical layer is still a logical communication because there is another, hidden layer, the transmission media, under the physical layer.
- Two devices are connected by a transmission medium (cable or air).

Physical Layer

- The transmission medium does not carry bits; it carries electrical or optical signals.
- The bits received in a frame from the data-link layer are transformed and sent through the transmission media.
- There are several protocols that transform a bit to a signal.

Data Link Layer

- An internet is made up of several links (LANs and WANs) connected by routers.
- There may be several overlapping sets of links that a datagram can travel from the host to the destination.
- The routers are responsible for choosing the best links.
- The link can be a wired LAN with a link-layer switch, a wireless LAN, a wired WAN, or a wireless WAN.
- The data-link layer is responsible for moving the packet through the link

Contd…

- TCP/IP does not define any specific protocol for the data-link layer.
- It supports all the standard and proprietary protocols.
- Any protocol that can take the datagram and carry it through the link suffices for the network layer.
- The data-link layer takes a datagram and encapsulates it in a packet called a frame.

Network Layer

- The **network layer** is responsible for creating a connection between **the source computer and the destination computer.**
- The communication at the network layer is host-to-host.
- There can be several routers from the source to the destination.
- The routers in the path are responsible for choosing the best route for each packet.

Contd…

- The routers do not need the application and transport layers
- The network layer in the Internet includes the main protocol, Internet Protocol (IP), that defines the format of the packet, called a datagram at the network layer.
- IP is also responsible for routing a packet from its source to its destination, which is achieved by each router forwarding the datagram to the next router in its path.

Contd…

- **IP is a connectionless protocol** that provides no flow control, no error control, and no congestion control services.
- The network layer also includes unicast (one-to-one) and multicast (one-to-many) routing protocols.
- A routing protocol does not take part in routing (it is the responsibility of IP), but it creates forwarding tables for routers to help them in the routing process.

Contd…

- The Internet Control Message Protocol (ICMP) helps IP to report some problems when routing a packet.
- The Internet Group Management Protocol (IGMP) is another protocol that helps IP in multitasking.
- The Dynamic Host Configuration Protocol (DHCP) helps IP to get the network-layer address for a host.
- The Address Resolution Protocol (ARP) is a protocol that helps IP to find a router when its network-layer address is given.

Transport Layer

- The logical connection at the transport layer is also end-to-end.
- The transport layer at the source host gets the message from the application layer, encapsulates it in a transport layer packet (called a segment or a user datagram in different protocols)
- Sends it, through the logical (imaginary) connection, to the transport layer at the destination host.
- The transport layer is responsible for giving services to the application layer: to get a message from an application program running on the source host and deliver it to the corresponding application program on the destination host.

Application Layer

- The logical connection between the two application layers is end-to-end.
- The two application layers exchange messages between each other as though there were a bridge between the two layers.
- Communication at the application layer is between two processes (two programs running at this layer).
- To communicate, a process sends a request to the other process and receives a response.
- Process-to-process communication is the duty of the application layer.

Contd…

- The Hypertext Transfer Protocol (HTTP) is a vehicle for accessing the World Wide Web (WWW).
- The Simple Mail Transfer Protocol (SMTP) is the main protocol used in electronic mail (e-mail) service.
- The File Transfer Protocol (FTP) is used for transferring files from one host to another.
- The Terminal Network (TELNET) and Secure Shell (SSH) are used for accessing a site remotely.

Contd…

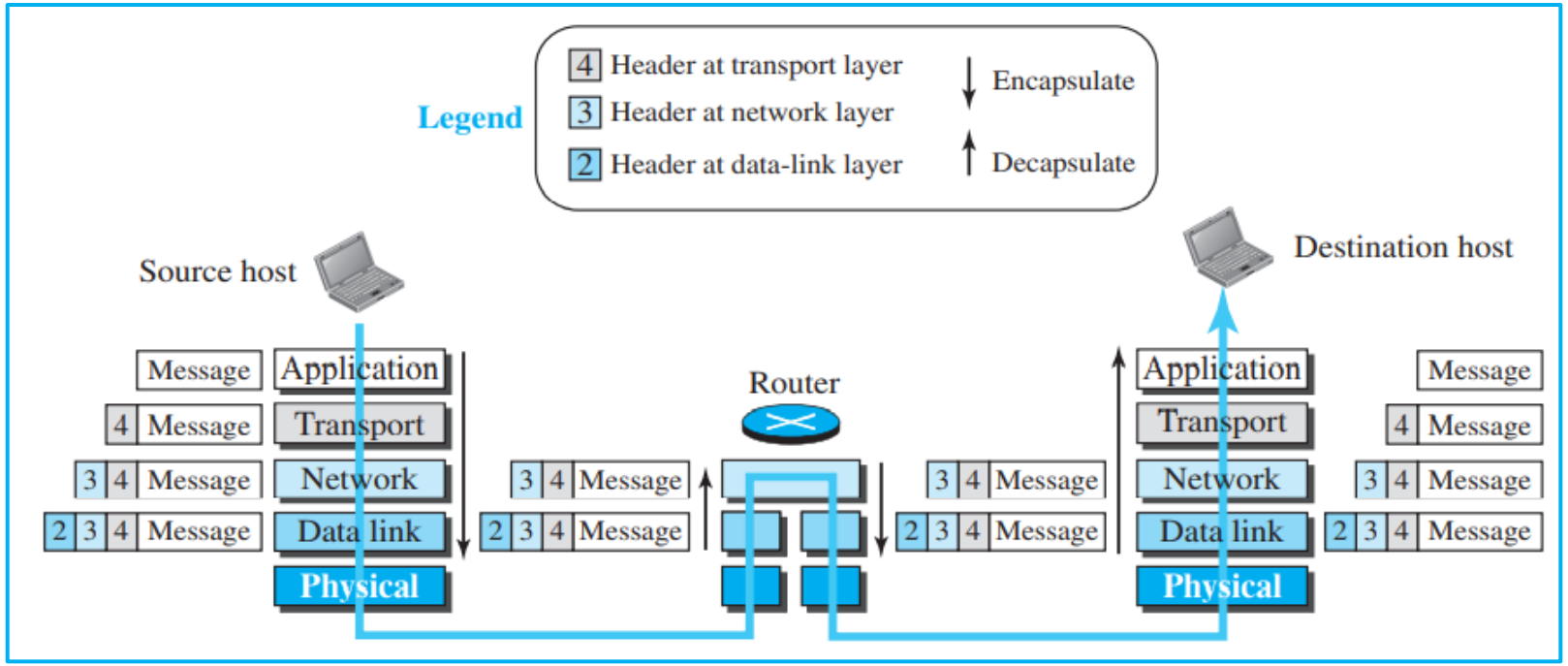
- The Simple Network Management Protocol (SNMP) is used by an administrator to manage the Internet at global and local levels.
- The Domain Name System (DNS) is used by other protocols to find the network-layer address of a computer.
- The Internet Group Management Protocol (IGMP) is used to collect membership in a group.

Encapsulation and Decapsulation

Contd…

- One of the important concepts in protocol layering in the Internet is encapsulation/decapsulation.
- The encapsulation in the source host, decapsulation in the destination host, and encapsulation and decapsulation in the router.

Encapsulation/ Decapsulation



Encapsulation at the Source Host

- At the application layer, the data to be exchanged is referred to as a message.
- A message normally does not contain any header or trailer, but if it does, we refer to the whole as the message.
- The message is passed to the transport layer.
- The transport layer takes the message as the payload, the load that the transport layer should take care of.

Contd...

- It adds the transport layer header to the payload, which contains the identifiers of the source and destination application programs that want to communicate plus some more information that is needed for the end-to-end delivery of the message, such as information needed for flow, error control, or congestion control.
- The result is the transport-layer packet, which is called the segment (in TCP) and the user datagram (in UDP).
- The transport layer then passes the packet to the network layer.

Contd...

- The network layer takes the transport-layer packet as data or payload and adds its own header to the payload.
- The header contains the addresses of the source and destination hosts and some more information used for error checking of the header, fragmentation information, and so on.
- The result is the network-layer packet, called a datagram.
- The network layer then passes the packet to the data-link layer.

Contd...

- The data-link layer takes the network-layer packet as data or payload and adds its own header, which contains the link-layer addresses of the host or the next hop (the router).
- The result is the link-layer packet, which is called a frame.
- The frame is passed to the physical layer for transmission.

Decapsulation and Encapsulation at the Router

- At the router, Both decapsulation and encapsulation because the router is connected to two or more links.

1. After the set of bits are delivered to the data-link layer, this layer decapsulates the datagram from the frame and passes it to the network layer.

2. The network layer only inspects the source and destination addresses in the datagram header and consults its forwarding table to find the next hop to which the datagram is to be delivered.

Decapsulation and Encapsulation at the Router

- The contents of the datagram should not be changed by the network layer in the router unless there is a need to fragment the datagram if it is too big to be passed through the next link.
- The datagram is then passed to the data-link layer of the next link.

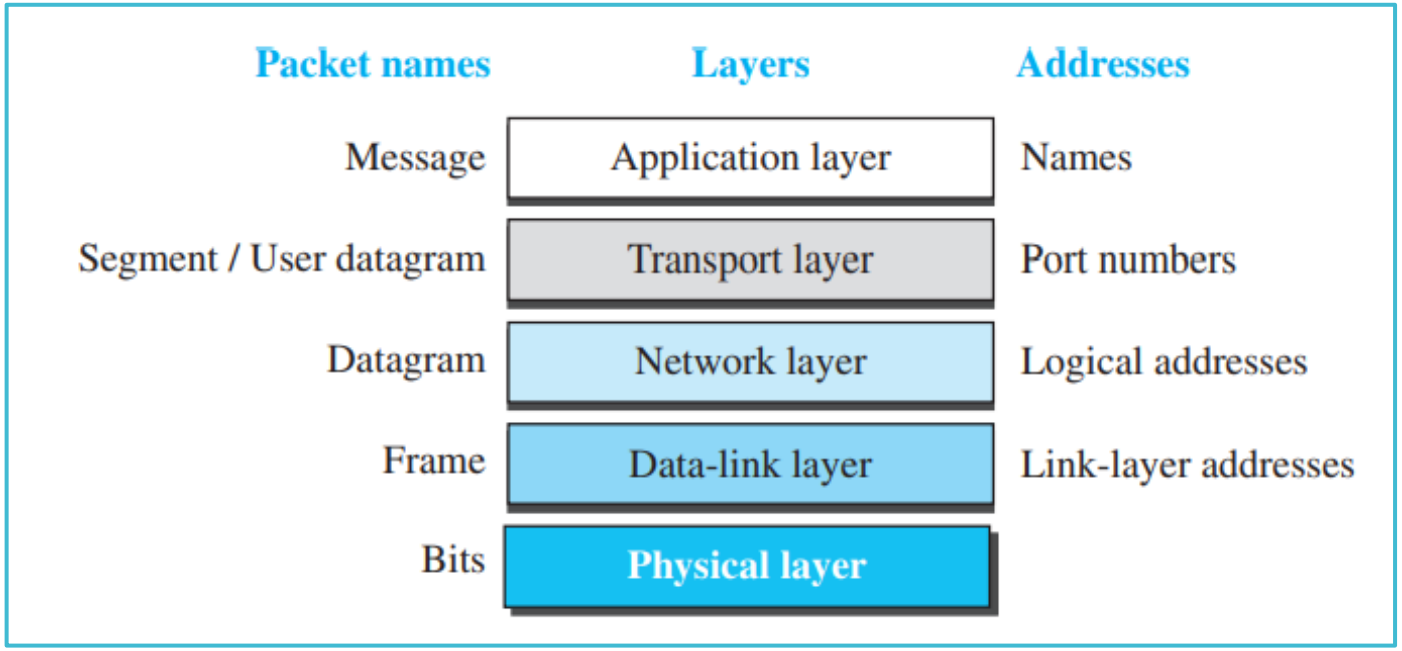
3. The data-link layer of the next link encapsulates the datagram in a frame and passes it to the physical layer for transmission.

Decapsulation at the Destination Host

- At the destination host, each layer only decapsulates the packet received, removes the payload, and delivers the payload to the next-higher layer protocol until the message reaches the application layer.
- It is necessary to say that decapsulation in the host involves error checking.

Addressing

- Any communication that involves two parties needs two addresses: source address and destination address.
- 5 pairs of addresses - one pair per layer
- only four because the physical layer does not need addresses



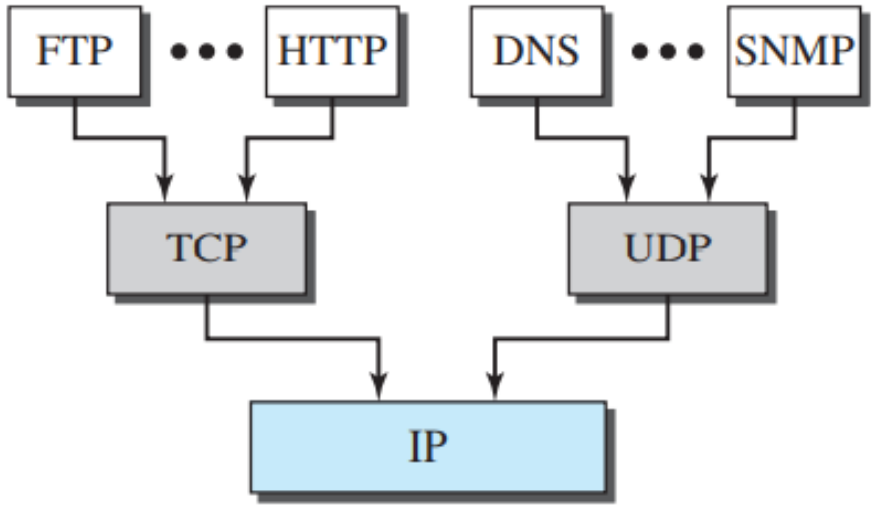
Contd...

- At the transport layer - addresses are called port numbers
- At the network-layer - the addresses are global, with the whole Internet as the scope
- Link-layer addresses - sometimes called MAC addresses, are locally defined addresses.

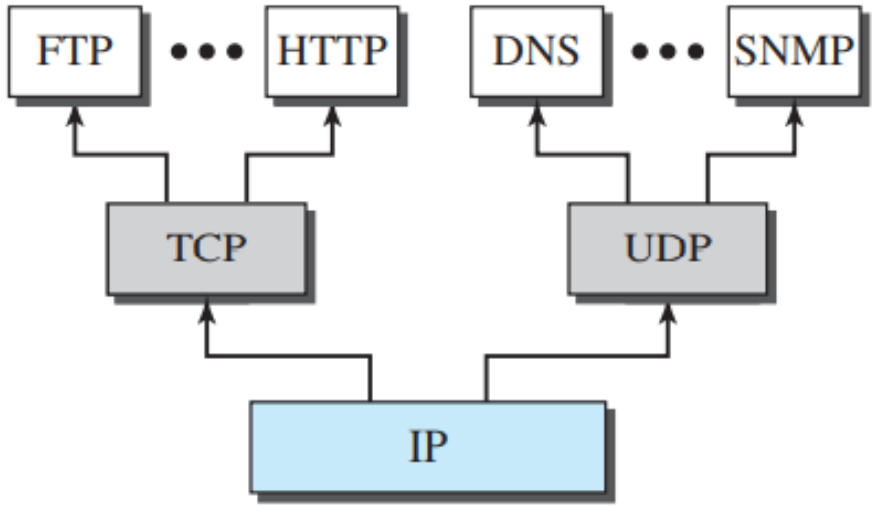
Multiplexing and Demultiplexing

- The TCP/IP protocol suite uses several protocols at some layers - multiplexing at the source and demultiplexing at the destination.
- Multiplexing means that a protocol at a layer can encapsulate a packet from several next-higher layer protocols (one at a time);
- Demultiplexing means that a protocol can decapsulate and deliver a packet to several next-higher layer protocols (one at a time).

Multiplexing and demultiplexing



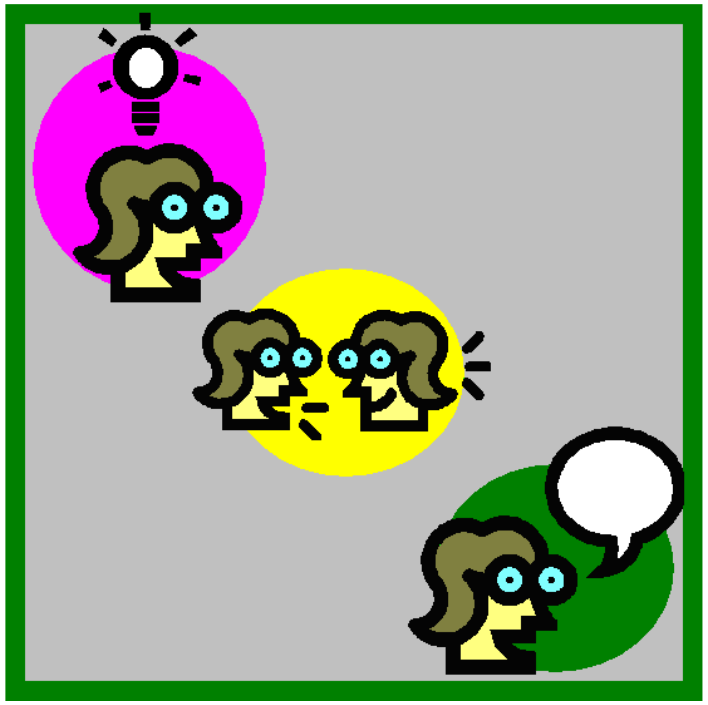
a. Multiplexing at source



b. Demultiplexing at destination

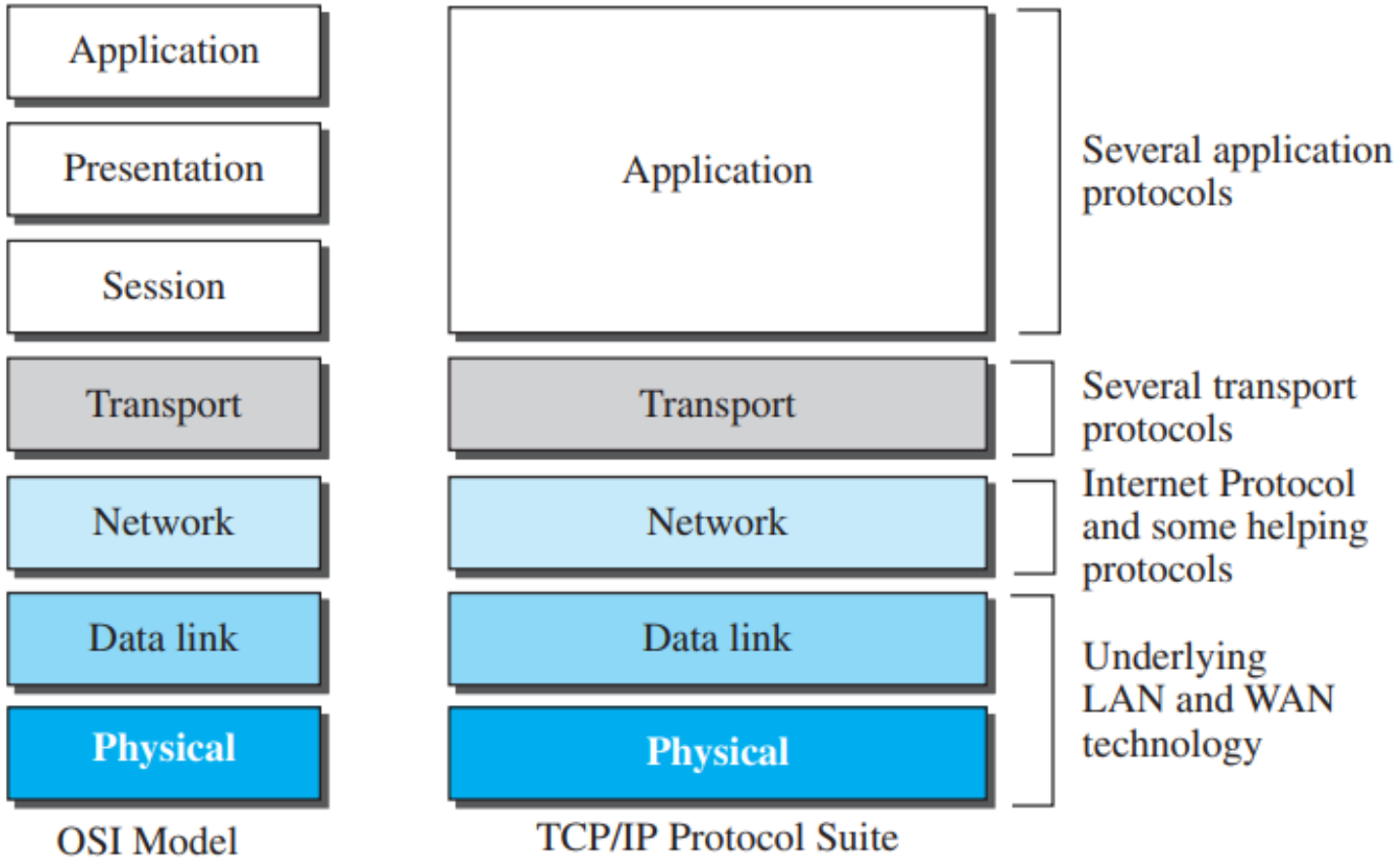
Activity – Think Pair Share

Comparison of OSI and TCP/IP Model



OSI versus TCP/IP

TCP/IP AND OSI MODEL



Contd…

- Two layers, session and presentation, are missing from the TCP/IP protocol suite.
- These two layers were not added to the TCP/IP protocol suite after the publication of the OSI model.
- The application layer in the suite is usually considered to be the combination of three layers in the OSI model.

Two reasons

- First, TCP/IP has more than one transport-layer protocol.
- Some of the functionalities of the session layer are available in some of the transport-layer protocols.

- Second, the application layer is not only one piece of software.
- Many applications can be developed at this layer.
- If some of the functionalities mentioned in the session and presentation layers are needed for a particular application, they can be included in the development of that piece of software.

Contd…

- The OSI model appeared after the TCP/IP protocol suite.
- **First,** OSI was completed when TCP/IP was fully in place and a lot of time and money had been spent on the suite; changing it would cost a lot.
- **Second,** some layers in the OSI model were never fully defined.

Contd…

- For example, although the services provided by the presentation and the session layers were listed in the document, actual protocols for these two layers were not fully defined, nor were they fully described, and the corresponding software was not fully developed.
- **Third,** when OSI was implemented by an organization in a different application, it did not show a high enough level of performance to entice the Internet authority to switch from the TCP/IP protocol suite to the OSI model.